

Shirley Zhang

xinyuezhang.shirley@gmail.com | (847) 964-0089 | LinkedIn: www.linkedin.com/in/xinyue-zhang-2292b225b

EDUCATION

Stanford University

Master of Science in Computer Science

Concentration: Information Management and Analytics

GPA: 3.97, Anticipated Graduation: December 2026

- **Relevant Courses:** Decision Making Under Uncertainty, Machine Learning, Mining Massive Datasets

Northwestern University

Bachelor of Science in Computer Science

Murphy Scholar, Summa Cum Laude

GPA: 3.99, Graduated June 2025

- **Relevant Courses:** Scalable Software Architecture, Introduction to Networking, Foundations of Security

SKILLS

Programming Languages: Python, SQL, C++, JavaScript, TypeScript, Java, C#

Cloud Infrastructure: Kubernetes, Docker, Linux, AWS (EKS, EC2, Lambda, SQS), AWS CDK, Git, Airflow

AI Infrastructure: PyTorch, TensorFlow, Langflow

Web & Application Development: React, Node.js, Express, REST APIs, HTML, CSS

EXPERIENCE

Software Engineering Intern | *Ironclad*

June 2026 – Present

- Architecting an end-to-end contract lifecycle platform that transforms chat-based procurement requests into structured contracting workflows, coordinating agent orchestration, information retrieval, decision-making, and downstream execution processes.
- Developing specialized contract classification and risk assessment agents that leverage retrieved contract metadata and business context to generate provenance-aware recommendations and structured outputs for redlining, execution, and review workflows.

Software Engineering Intern | *XXX*

January 2026 – June 2026

- Architected an enterprise AI query platform enabling natural language access to structured databases and enterprise documents through a unified Query Router that classified user intent and orchestrated Text-to-SQL, RAG, and hybrid retrieval pipelines with intelligent fallback across heterogeneous data sources.
- Built production Text-to-SQL and citation-grounded RAG systems with automated schema selection for 100+ table databases, grounded retrieval, and LLM guardrails, achieving XX% query accuracy on XXX+ internal benchmark cases.
- Optimized production performance through multi-layer Redis caching, multi-tenant security, and request auditing, reducing LLM API calls by XX% while achieving P95 latency < XXX ms.

Software Engineering Intern | *PwC*

June 2025 – August 2025

- Designed a multi-agent enterprise knowledge assistant using Langflow orchestration, implementing supervisor, retrieval, and response-generation workflows that routed employee questions through Azure AI Search and citation-grounded RAG pipelines over internal documentation, with architecture designed to support PwC's global workforce of 300K+ employees.
- Built automated evaluation pipelines for a citation-grounded RAG system using a human-curated benchmark spanning 50+ internal documents and hundreds of representative queries, and performing manual edge-case testing and prompt refinement to improve retrieval recall, answer grounding, and hallucination detection.
- Refactored backend storage and frontend rendering architecture to resolve hierarchical data consistency issues and cascading UI failures, improving platform reliability and ensuring accurate navigation of documentation across newly restructured organizational workflows.

Data Analyst Intern | *Tesla*

March 2025 – June 2025

- Architected a fleet-scale brake deterioration detection platform processing telemetry from 1M+ Model 3 vehicles, designing data ingestion, feature engineering, model inference, and visualizations to identify vehicles at risk of brake fluid leaks.
- Built a Spark-based serving pipeline that automated daily fleet health analysis, generating vehicle-level risk scores and surfacing high-priority cases through internal dashboards used by chassis and service teams.
- Delivered a 96.6% precision model validated on serviced vehicles, identifying approximately 25% of deteriorating brake systems up to two months before service and enabling transition from manual investigations to continuous fleet monitoring.

PROJECTS

MuseLab: a Multi-Agent Literary Intelligence Platform

March 2026 – June 2026

- Designed a full-stack literary analysis platform using React, TypeScript, Express, and PostgreSQL, implementing literary-analysis services, semantic visualization engines, and built evaluation systems to support end-to-end revision workflows.
- Developed a 14-agent reasoning architecture combining literary-analysis agents with human-in-the-loop evaluation agents, enabling autonomous critique of both user manuscripts and the platform itself through specialized workflows for interpretation, software quality, UX evaluation, and design review.